

AIRBNB INTELLIGENCE

Unlocking Business Value from Listing Descriptions

Data Visualization & Text Analytics | Spring 2026

Submitted: March 26, 2026

1. Executive Summary

This analysis covers 5555 Airbnb listings across nine markets, drawing on over 365,000 tokens from host listing description. Average host response rate is 94.5 percent, median nightly price is \$155, and total reviews stand at 101,000. Review scores vary, with the United States highest at 94%, followed by Australia and Canada at 93, Spain at 91, and China at 89. This entails that higher-rated markets set clear expectations before making aspirational claims, suggesting that precise, honest descriptions improve guest trust and ratings and strongly recommended.

Pricing varies by property type, with entire homes at \$140, private rooms at \$55, shared rooms at \$51, and resorts or chalets exceeding \$300. This means that pricing reflects different guest segments hence it is highly recommended that property type and description strategy be aligned to justify value and attract the right audience.

Text analytics reveals that US hosts emphasize parking and neighborhood context, while Spanish hosts highlight transit and walking times. It is therefore recommended that Airbnb tailor its messaging to local guest priorities to maximize engagement and conversion. Positive language outpaces negative ones roughly two to one, with words like *enjoy*, *quiet*, and *cozy* boosting appeal, and it is recommended that negatives words like *noise* require reframing. Topic modeling revealed four themes: location, experience, amenities, and local context which are driving discoverability, bookings, and guest satisfaction. A quality scoring system to guide hosts to incorporate these themes is recommended, turning listing optimization into a measurable driver of revenue growth.

2. Visualizations

Country-Level Review Score: a grouped bar chart to uncover how review scores stands across the five most active markets: The United States leads in satisfaction, while China trails by five

points despite its size, a gap which reflects cultural differences in review practices, translation challenges in listings, or even unmet expectations around Western-style amenities.

Listing Capacity vs. Bed Availability: a scatter analyzing guest capacity versus total beds, revealing that most listings cluster between 1–5 guests and 1–5 beds, consistent with the platform average of one bedroom and two beds. A smaller segment accommodates 15–20 guests with 10 or more beds. High-capacity listings present opportunities; a targeted group-stays strategy could increase visibility, bookings, and revenue.

Price Variation by Property Type: a bar chart tracking how property type ranks against the median price, with Resorts (\$350+) and Chalets (\$250+) at the top, and Apartments. Property type drives pricing more than market, therefore hosts can earn revenue by enhancing features, targeting premium categories, while Airbnb can grow bookings by expanding resort and chalet inventory.

Average Price by Room Type: a bar chart to determine price bands for three room types: Entire Home/Apartment (\$140 median), Private Room (\$55), and Shared Room (\$51). The minimal highlight Private and Shared rooms highlights an opportunity to reframe Shared listings, while the 2.7× premium for Entire Homes confirms privacy as the main value driver.

3. Dashboard Description

The Power BI dashboard provides a unified, interactive view of Airbnb's global listings through six panels. A KPI banner offers a quick overview of key portfolio metrics. Filters on country and property type allow users to explore regional and segment-level performance. Visualizations include a review score bar chart, a scatter plot of guest capacity versus beds to highlight large-capacity listings, and side-by-side pricing panels by property and room type, giving teams a comprehensive view of performance, supply distribution, and revenue opportunities.

4. Text Mining: Findings & Business Insights

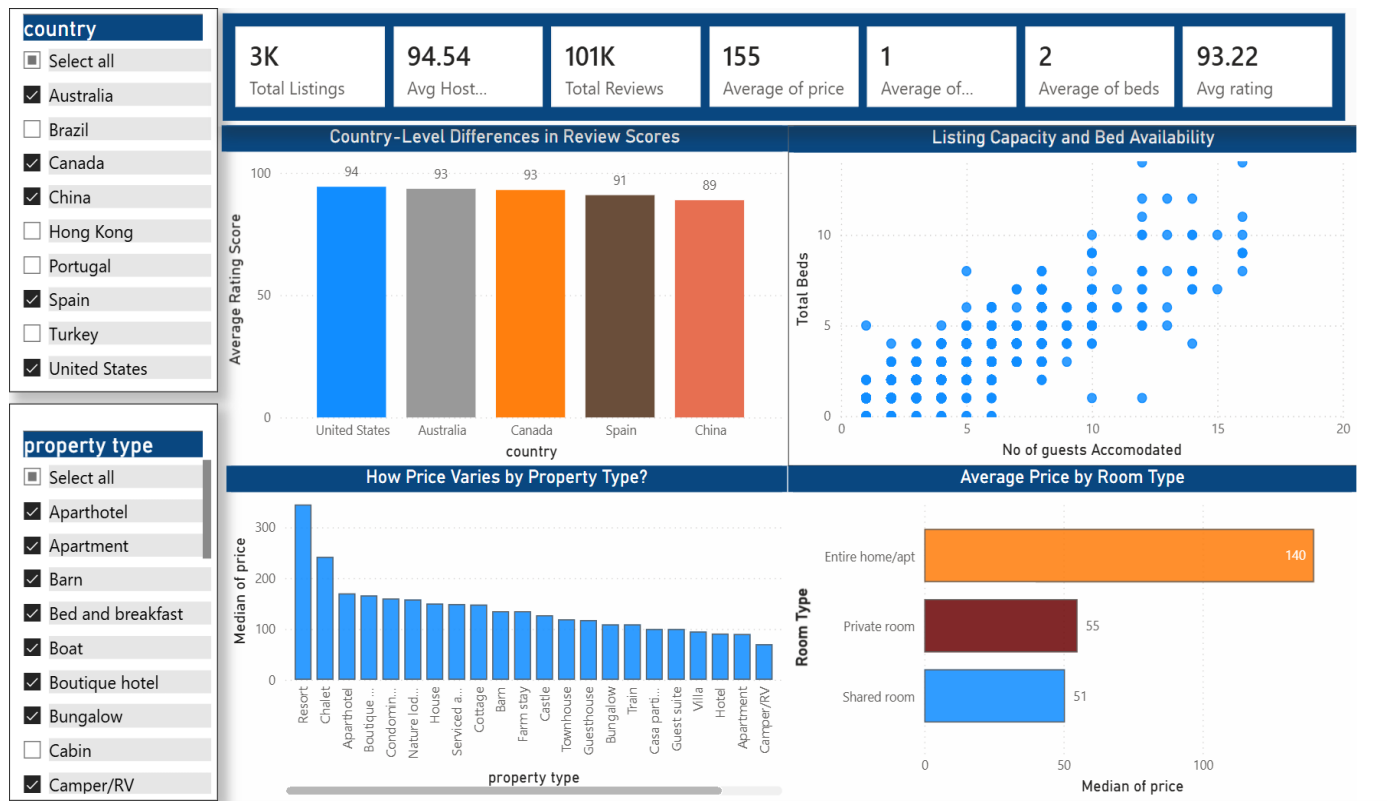
Word frequency and bigram analysis reveals that hosts emphasize layout and location, with terms like "apartment," "bedroom," and "walk" dominating. Moreso, bigrams like "minute walk" reveal that quantified walkability is the most persuasive language in descriptions, though phrasing varies by market, for instance "metro station" for dense cities. Bigram like "air condition" tops the list reflecting regional expectations. Airbnb should provide host coaching on proximity and key amenities to improve search ranking and guest engagement.

Bing lexicon Sentiment analysis shows positive tokens consistently outnumber negatives by 50% across nine markets, indicating hosts treat listings as marketing copy. Top positive words like *enjoy, comfort, spacious, clean, modern, and cozy* reflect guest priorities such as tranquility, space, and cleanliness. Frequent negative words such as *noise, smoke, complex, split, sink* often misrepresent neutral features in context. Airbnb should provide a language listing guide that reframes ambiguous negatives like *complex* and emphasizes comfort and spaciousness to improve search ranking and guest conversion.

Topic modelling using LDA was applied to the text and revealed four thematic clusters in how hosts frame descriptions with **Topic 1** emphasizes apartment features and amenities, highlighting proximity to restaurants, shops, and transit. **Topic 2** focuses on guest experience and host hospitality, using emotionally resonant language to sell a feeling, not just a space. **Topic 3** covers functional aspects such as bedrooms, kitchens, bathrooms, and Wi-Fi, reflecting baseline guest expectations. **Topic 4** captures neighborhood and local attractions, with non-English terms indicating listings targeting domestic audiences in multilingual markets. Listings that combine location, experience, amenities, and local authenticity best meet guest needs, and Airbnb coaching with a scoring rubric can guide higher-performing descriptions.

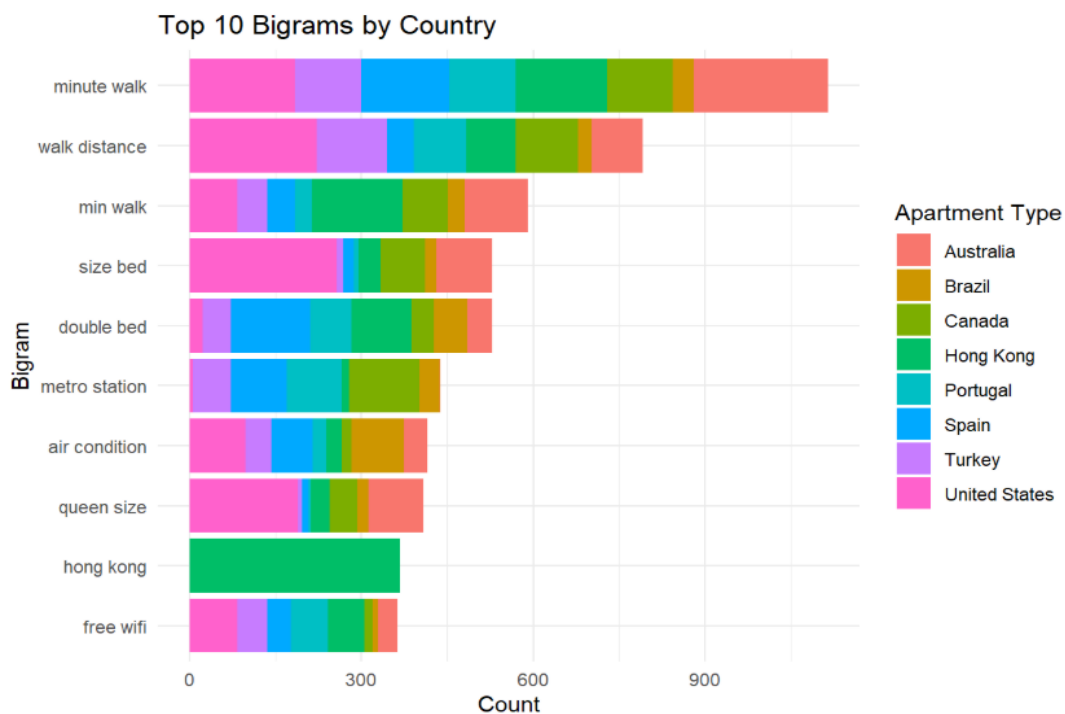
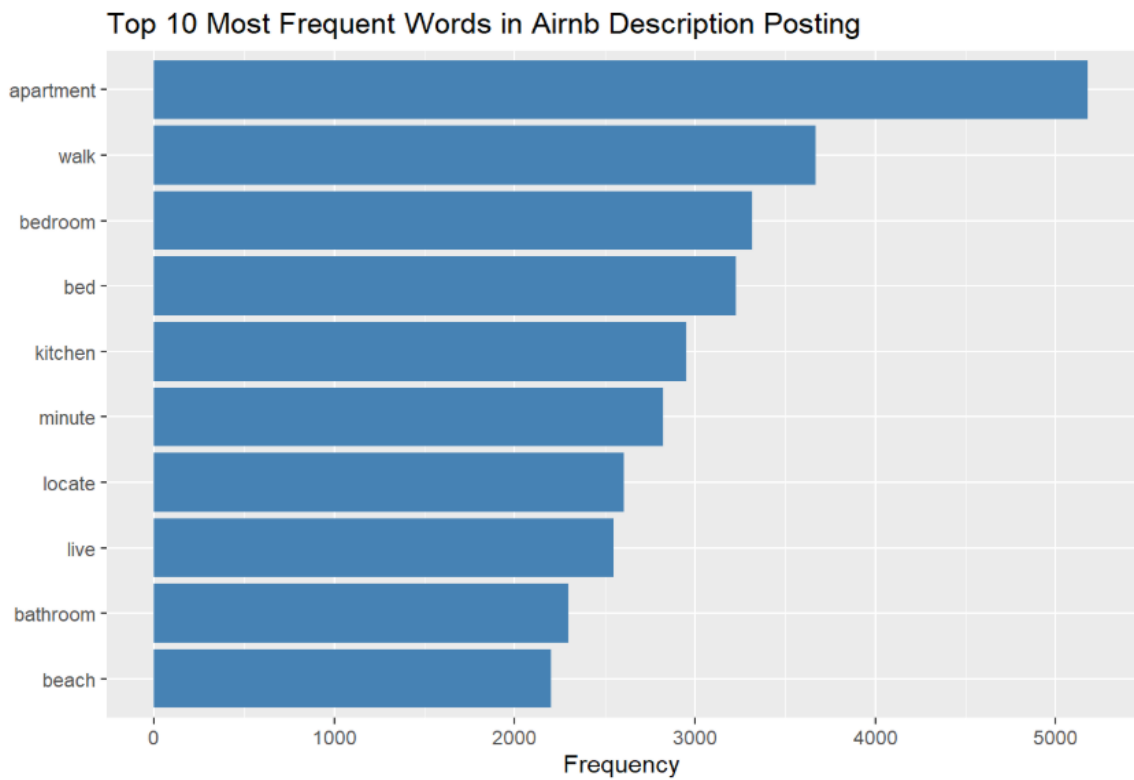
Appendix A: Dashboard Screenshots

The screenshot below showcases the complete Power BI dashboard developed for this analysis, including enhanced tooltips that provide additional context and insights on hover. It features a KPI banner for quick performance insights, along with interactive filters for country and property type to enable dynamic exploration. The dashboard also includes a review score comparison chart, a scatter plot illustrating listing capacity, and pricing visualizations segmented by both property type and room type.

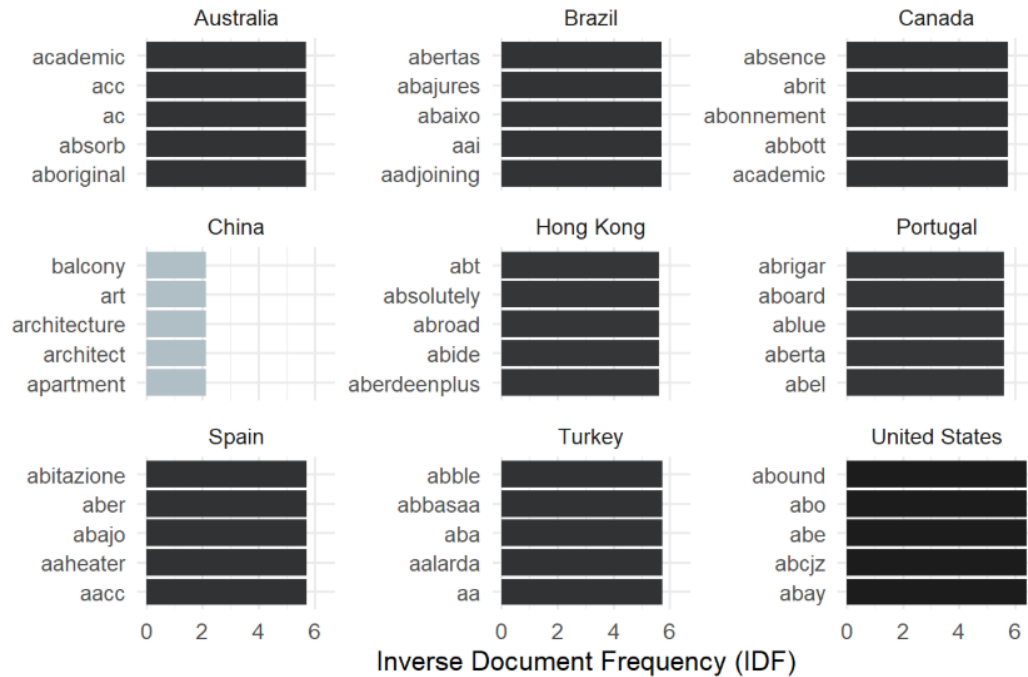


Appendix B:

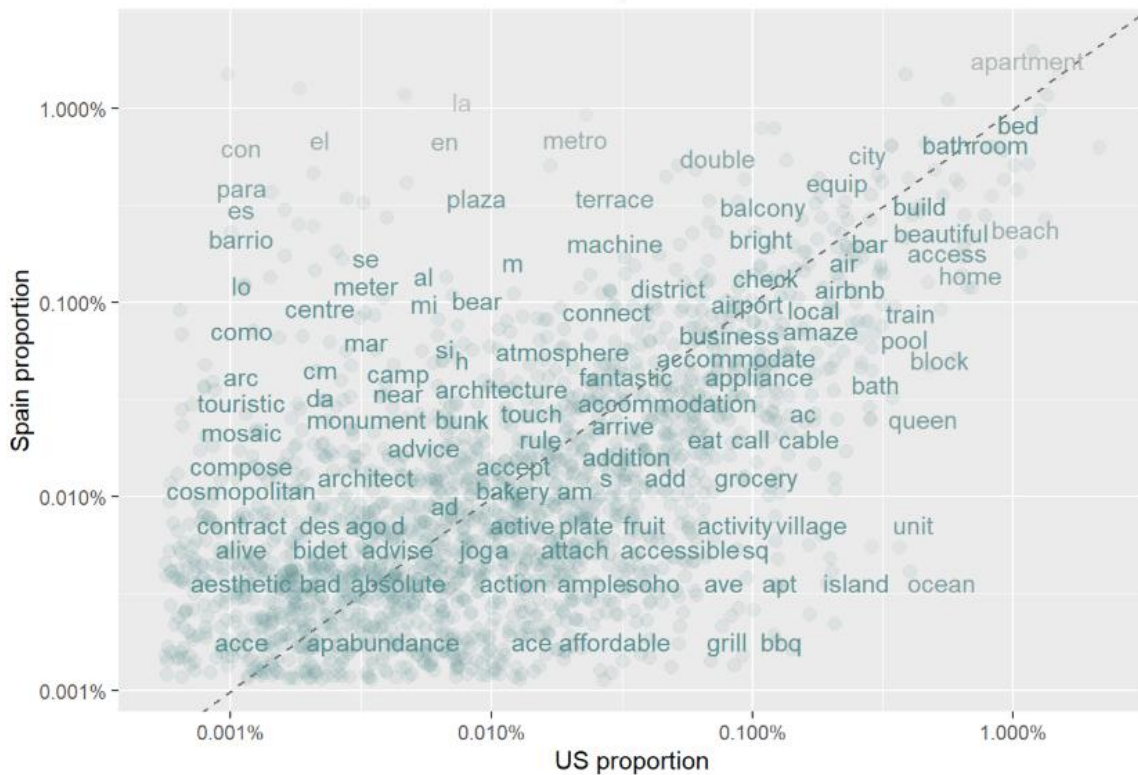
Visual Assets



Top 5 Most Rare Words in Airbnb Descriptions by Country



Word Correlations: United States vs Spain



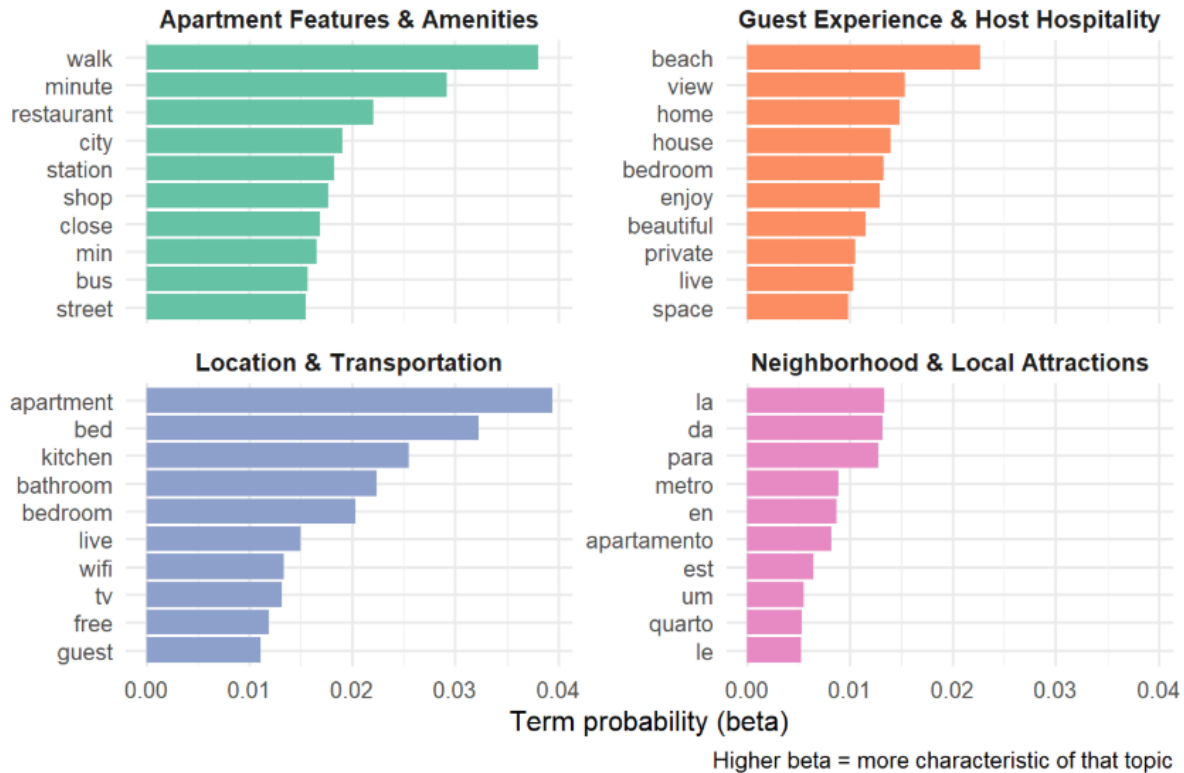
Stacked bar chart showing Word Count (Y-axis, 0 to 20,000+) by Sentiment (X-axis: negative, positive) and Country (Legend).

The chart displays word counts for negative and positive sentiments across nine countries. The positive sentiment bar is significantly larger than the negative sentiment bar. The countries included in the legend are Australia, Brazil, Canada, China, Hong Kong, Portugal, Spain, Turkey, and United States.

Sentiment	Australia	Brazil	Canada	China	Hong Kong	Portugal	Spain	Turkey	United States
negative	~1,000	~500	~500	~500	~500	~500	~500	~500	~1,000
positive	~10,000	~2,000	~3,000	~2,000	~2,000	~2,000	~2,000	~2,000	~9,000

[illegible]

LDA topic modeling - Airbnb Listing Descriptions



R code used to structure the text data

```
# Loading Data from MongoDB
```

```
## Setting up the connection to get the data from MongoDB
```

```
MONGO_URI <-
```

```
"mongodb+srv://hmupfumi_db_user:pfWv1ZuX0L8Gc5fu@cluster0.2opjt6h.mongodb.net/ampl  
e_airbnb?retryWrites=true&w=majority"
```

```
DB_NAME <- "sample_airbnb"
```

```
## Loading Airbn data
```

```
#Fetch the data from Airbnb
```

```
con_airbnb <- mongo(
```

```

collection = "listingsAndReviews",
db         = DB_NAME,
url        = MONGO_URI
)

```

```

airbnb_raw <- con_airbnb$find(
  query = '{}',
  fields = '{"_id": 0}'
)

```

#creating a dataframe with description and location information

```

airbnb_df <- data.frame(
  country      = airbnb_raw$address$country,
  property_type = airbnb_raw$property_type,
  description   = airbnb_raw$description,
  review_rating = airbnb_raw$review_score$review_scores_rating,
  price         = airbnb_raw$price
)%>%
  mutate(doc_id = row_number())

```

Text Preprocessing

Text Cleaning Pipeline

```

clean_text <- function(text) {
  text %>%
    str_to_lower() %>%                # convert to lowercase
    str_remove_all("https?:/\\S+|www\\.\\S+") %>% # Remove URLs
    str_remove_all("@\\w+|#\\w+") %>%      # mentions & hashtags
    str_replace_all("n't", " not") %>%    # Replace contraction n't with not
    str_replace_all("'re", " are") %>%    # Replace contraction 're with are
    str_replace_all("'ve", " have") %>%   # Replace contraction 've with have
    str_replace_all("'ll", " will") %>%   # Replace contraction 'll with will
    str_replace_all("won't", "will not") %>% # Replace contraction won't with will not
    str_replace_all("can't", "can not") %>% # Replace contraction cant with can not

```



```

!word2 %in% stop_words$word,
!str_detect(word1, "^\\d+$"),          # remove numbers
!str_detect(word2, "^\\d+$")
) %>%
mutate(
  word1 = lemmatize_words(word1),      # lemmatize
  word2 = lemmatize_words(word2)
) %>%
  unite(bigram_lemma, word1, word2, sep = " ") %>%      # combine back into
single column
dplyr::select(doc_id,country,bigram_lemma)

```

*Quick summary*

```

cat("Total bigrams: ", nrow(airbnb_bigrams), "\n")
cat("Unique bigrams: ", n_distinct(airbnb_bigrams$bigram_lemma), "\n")

```